Taut fillings

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Abstract

Let σ be a simplicial triangulation of the 2-sphere, X the associated integral 2-cycle. A filling of X is an integral 3-chain M with $\partial M = X$; a taut filling is one with minimal L_1-norm. We show that any taut filling arises from an extension of σ to a simplicial complex homeomorphic to the 3-ball. The filling is *clean*: it has no repeated tetrahedron, and its support complex is a clean simplicial complex. This support complex is shellable and flag: every clique in its 1-skeleton occurs as a simplex. The key to the proof is the general fact that any taut filling of an n-cycle splits under disjoint union, connected sum, and more generally what we call almost disjoint union, where summands are supported on sets that overlap in at most $n+1$ vertices.

1 Overview

Let $\Delta = \Delta(\Omega)$ be the $|\Omega|-1$-simplex with vertices Ω, viewed as the abstract simplicial complex consisting of all subsets of Ω. Let $C_n = C_n(\Omega) = C_n(\Delta(\Omega), \mathbf{Z})$ and $Z_n = Z_n(\Omega) = Z_n(\Delta(\Omega), \mathbf{Z})$ be its integral n-chains and n-cycles. A *filling* of an n-cycle $X \in Z_n$ is any $n+1$-chain $M \in C_{n+1}$ with $\partial M = X$. (Here and throughout we assume $n \geq 1$.) Let $\mathrm{Zvol}(X)$ be the minimum L_1-norm of a filling of X. Call $M \in C_{n+1}$ *taut* if it is an optimal filling of its boundary, i.e., if $|M| = \mathrm{Zvol}(\partial M)$.

For $X \in C_n$ let $\mathrm{Vert}(X)$ be the set of all the vertices of all the n-simplices to which X assigns non-0 weight. For any taut filling M of X we have

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$\mathrm{Vert}(M) = \mathrm{Vert}(X)$. (Cf. Proposition 4 below.) This is why we write C_n and Z_n without reference to Ω.

Call an n-cycle $X + Y \in Z_n$ an *almost disjoint union* if

$$|\mathrm{Vert}(X) \cap \mathrm{Vert}(Y)| \leq n + 1.$$

This notion generalizes both disjoint union and connected sum along an n-simplex. Ellison [2] showed the Zvol adds under almost disjoint union: $\mathrm{Zvol}(X + Y) = \mathrm{Zvol}(X) + \mathrm{Zvol}(Y)$. This means that *some* taut filling M of X splits into a sum $M = M_X + M_Y$ of taut fillings of X, Y. Here we amplify Ellison's Corollary 2 to show (Theorem 1) that when $n \geq 2$, *any* taut filling of $X + Y$ splits.

In place of integral chains we can use chains with coefficients in $\mathbf{Q}$ or $\mathbf{R}$. Evidently Rvol $\leq$ Qvol. But computing Rvol is a linear programming problem with rational coefficients, so in fact Qvol = Rvol. Ellison used LP duality to prove that Qvol adds under almost disjoint union. Now for $n \geq 2$ we get the stronger result (Corollary 2) that taut $\mathbf{Q}$-fillings split: Multiplying a taut $\mathbf{Q}$-filling M of X by a common denominator q yields a taut $\mathbf{Z}$-filling qM of qX: Split the $\mathbf{Z}$-filling qM and divide by q to get a splitting of the $\mathbf{Q}$-filling M.

Now let σ be a simplicial triangulation of S^2; let $X(\sigma) \in Z_2(\Delta(\mathrm{Vert}(\sigma)), \mathbf{Z})$ be either one of the 2-cycles that arises from σ by orienting its 2-simplices; and write $\mathrm{Zvol}(\sigma) = \mathrm{Zvol}(X(\sigma))$. Let $\mathrm{tetvol}(\sigma)$ be the minimum number of 3-simplices required to extend σ to a simplicial triangulation τ of B^3. We will show that

$$\mathrm{Zvol}(\sigma) = \mathrm{tetvol}(\sigma).$$

Certainly Zvol $\leq$ tetvol, because any extension τ induces a filling of $X(\sigma)$. Theorem 2 states that any taut filling of $X(\sigma)$ is clean and that its support complex $K(M)$ triangulates B^3. The proof of Theorem 2 relies on Theorem 1.

We do not know what happens for spheres of dimension 3 or greater.

2 Background

Our interest in Zvol stems from the work of Sleator, Tarjan, and Thurston [5]. They observed that coning from a vertex of maximal degree shows that a v-vertex triangulation σ of S^2 has tetvol $\leq 2v - 4 - \mathrm{maxdeg}(\sigma)$. If $v \geq 13$

maxdeg(σ) ≥ 6, so tetvol $\leq 2v - 10$. They produced examples for which tetvol $= 2v - 10$, provided v is larger than some unspecified bound, and conjectured that such examples should exist for any $v \geq 13$. Their approach was to realize σ as an ideal hyperbolic polyhedron with volume $2vV_0 - O(\log(v))$, where V_0 is the volume of an equilateral ideal tetrahedron, which is maximal. This implies tetvol $\geq 2v - O(\log(v))$. From there they bootstrapped their way up to showing that tetvol $= 2v - 10$. They suggested [5, p. 697] that in fact Qvol $= 2v - 10$, which would immediately imply tetvol $= 2v - 10$. Mathieu and Thurston [3] produced a class of examples, different from the examples of STT, for which they could show that Qvol $= 2v - 10$, still under the assumption that v is sufficiently large. In [1] we produced examples for any $v \geq 13$ with Qvol $= 2v - 10$, confirming the conjecture of Sleator, Tarjan, and Thurston. Theorem 2 here tells us that whenever we have Qvol $= 2v - 10$, or just Zvol $= 2v - 10$, any taut filling must arise from a triangulation of the ball.

About Qvol versus Zvol. In [1] we described triangulations of S^2 with Qvol $<$ Zvol. In all such examples that we know of with maxdeg ≤ 6, the gap between Qvol and Zvol is < 1, so that $\lceil \text{Qvol} \rceil =$ Zvol. Since Qvol and Zvol add under connected sum, the gap between them can be arbitrarily large (Ellison [2]), but taking the connected sum pushes maxdeg above 6.

3 Taut fillings

We can think of an n-chain $X \in C_n$ as a multiset of non-cancelling oriented simplices:

$$X = \sum_{t \in X} t.$$

In this sum t denotes an oriented simplex

$$t = [x_0, \dots, x_n] = [x_{\pi(0)}, \dots, x_{\pi(n)}], \quad \pi \text{ an even permutation},$$

and each unoriented simplex contributes a number of terms corresponding to its multiplicity.

The size of M as a multiset is its L_1-norm $|M|$. Write $U \subset M$ if U is a sub-multiset of M. This happens just when $|M| = |U| + |M - U|$.

Proposition 1. *If M is taut and $U \subset M$ then U is taut.*

Proof. If U is not taut, there is a smaller filling V of ∂U. Take $M' = M - U + V$. In terms of multisets, this means that we substitute V for U, and then do any required cancellation of oppositely oriented simplices of $M - U$ and V. We have

$$\partial M' = \partial M - \partial U + \partial V = \partial M$$

and

$$|M'| \leq |M - U| + |V| = |M| - |U| + |V| < |M|,$$

contradiction. $\square$

Corollary 1. *A taut chain contains no nonzero closed subchain: if M is taut, $U \subset M$, and $\partial U = 0$, then $U = 0$.*

Proof. By Proposition 1, U is taut, so $|U| = \mathrm{Zvol}(\partial U) = \mathrm{Zvol}(0) = 0$, whence $U = 0$. $\square$

4 Coning

For $x \in \Omega$, $U \in C_n$ let

$$\mathrm{nbhd}(x, U) = \sum_{t \in U : x \in \mathrm{Vert}(t)} t,$$

$$\deg(x, U) = |\mathrm{nbhd}(x, U)|,$$

$$\mathrm{maxdeg}(U) = \max_x \deg(x, U).$$

The *cone from x to U* is the $(n+1)$-chain

$$\mathrm{cone}(x, U) = \sum_{t \in U} \mathrm{adj}(x, t),$$

consisting of all the non-trivial oriented $(n+1)$-simplices $\mathrm{adj}(x, t) = [x x_0 \ldots x_n]$ obtained by adjoining x to $t = [x_0 \ldots x_n] \in U$. If $x \in \mathrm{Vert}(t)$ then t does not contribute to the sum, so $|\mathrm{cone}(x, U)| = |U| - \deg(x, U)$.

If U is closed then $\partial\,\mathrm{cone}(x, U) = U$, so

Proposition 2. *For any $X \in Z_n$*

$$\mathrm{Zvol}(X) \leq |X| - \mathrm{maxdeg}(X) \qquad \square$$

If $U \in Z_n$ and $x \notin \mathrm{Vert}(U)$ then $\deg(x, U) = 0$ and $|\mathrm{cone}(x, U)| = |U|$. In this case we call $\mathrm{cone}(x, U)$ a *complete cone*. By Proposition 2 a non-trivial complete cone is not taut. (Equivalently, instead of coning from $x \notin U$, one could have coned from some $x \in U$.) Thus:

Proposition 3. *If M is taut it contains no non-trivial complete cone.* $\square$

Call $x \in \mathrm{Vert}(M) \setminus \mathrm{Vert}(\partial M)$ an *internal vertex* of M. If x is internal to M then $\mathrm{nbhd}(x, M)$ is a complete cone, so:

Proposition 4. *If M is taut then it has no internal vertices.* $\square$

5 Almost disjoint unions

Recall that if $X, Y \in Z_n$ we call $X + Y$ an almost disjoint union if

$$|\mathrm{Vert}(X) \cap \mathrm{Vert}(Y)| \leq n + 1.$$

The most interesting special case is a connected sum, where $t = [c_0, c_1, \ldots, c_n]$ occurs once in X and $-t = [c_1, c_0, \ldots, c_n]$ occurs once in Y. For example, if $n = 1$ with X a cycle of length p and Y a cycle of length q the connected sum $X+Y$ is a cycle of length $p+q-2$. Here Zvol adds, because $\mathrm{Zvol}(X) = p-2$, $\mathrm{Zvol}(Y) = q - 2$, and

$$\mathrm{Zvol}(X + Y) = (p + q - 2) - 2 = \mathrm{Zvol}(X) + \mathrm{Zvol}(Y).$$

But not every filling of $X + Y$ splits, because a taut filling may contain any 2-simplex with vertices in $\mathrm{Vert}(X + Y)$.

We want to show that fillings do split when $n \geq 2$.

For $A \subset \Omega$, $p \in A$ define

$$\pi_{A,p} : \Omega \to A$$

$$\pi_{A,p}(x) = x \textbf{ if } x \in A \textbf{ else } p$$

and let

$$K_*(A, p) : C_*(\Omega) \to C_*(A)$$

be the induced chain map. This is a projection of $C_*(\Omega)$ onto $C_*(A)$.

Let A, B be finite vertex sets sharing the vertices $C = A \cap B$, and let $c = |C|$. Observe that $C_c(C)$ and $Z_{c-1}(C)$ are trivial, because $\Delta(C)$ contains no c-simplex, and (up to orientation) only one $c-1$-simplex, with nothing to cancel its boundary.

Let

$$C_*(A, B) = C_*(A) \oplus C_*(B).$$

For $p, q \in C$ define the mapping

$$g_*(p, q) = K_*(A, p) \oplus K_*(B, q) : C_*(A \cup B) \to C_*(A, B).$$

Proposition 5. *Let $(X, Y) \in C_n(A, B)$. If $|A \cap B| \leq n$ we can recover (X, Y) from $X + Y$. If $(X, Y) \in Z_n(A, B)$ this holds also when $|A \cap B| = n + 1$.*

Proof. Let $C = A \cap B$. We can assume $|C| \geq 1$. (Add a brand new point to C if necessary.) We claim that for any $p, q \in C$ (not necessarily distinct) we have

$$g_n(p, q)(X + Y) = (X, Y).$$

Indeed,

$$K_n(A, p)(X + Y) = X + K_n(A, p)(Y).$$

The second term belongs to $C_n(C)$, which is trivial when $|C| \leq n$. If $Y \in Z_n(B)$ the second term belongs to $Z_n(C)$, which is trivial when $|C| \leq n + 1$. $\square$

Theorem 1. *If $|A \cap B| \leq n + 1$, for all $(X, Y) \in Z_n(A, B)$ we have*

$$\mathrm{Zvol}(X + Y) = \mathrm{Zvol}(X) + \mathrm{Zvol}(Y).$$

And as long as $n \geq 2$, for any $M \in \mathrm{taut}(X + Y)$ we have $M = M_X + M_Y$ with $M_X \in \mathrm{taut}(X)$, $M_Y \in \mathrm{taut}(Y)$.

Note. Ellison [2] proved the additivity of Zvol. What's new here is the splitting of taut fillings when $n \geq 2$. This result resembles Theorem 6.2 of Pournin and Wang [4] about flip paths. Like theirs, our proof uses a variation on the normalization technique of STT [5, Lemma 7]. It would be nice to fit these results under one roof.

Proof. Again let $C = A \cap B$. We can assume $|C| = n + 1$, as this is the hardest case. And we might as well go ahead and take $n = 2$, $|C| = 3$, as this case illustrates all the issues.

Take any $M \in \mathrm{taut}(X + Y) \subset C_3(A \cup B)$. Pick distinct points $p, q \in C$, and let

$$(M_X, M_Y) = g_3(p, q)(M).$$

(For now we suppress the dependence of M_X, M_Y on p, q.)

Because $g_*(p,q)$ is a chain map we have $\partial_3 M_X = X$, $\partial_3 M_Y = Y$:

$$(\partial_3 M_X, \partial_3 M_Y) = \partial_3(g_3(p,q)(M)) = g_2(p,q)(\partial_3 M) = g_2(p,q)(X+Y) = (X,Y).$$

We want to show that under the map $g_3(p,q)$ any $t \in M$ dies either in M_X or M_Y, because then additivity of Zvol follows from

$$\mathrm{Zvol}(X+Y) = |M| \geq |M_X| + |M_Y| \geq \mathrm{Zvol}(X) + \mathrm{Zvol}(Y) \geq \mathrm{Zvol}(X+Y).$$

Let's call a 3-simplex a 'tet' for short. Say that a tet $t \in M$ has type $CCXY$ if $t = \pm[c_1 c_2 x_1 y_1]$ for $c_1, c_2 \in C$, $x_1 \in A \setminus C$, $y_1 \in B \setminus C$. Similarly for types $XXXX$, $CXXX$, $CXXY$, $XXXY$, etc. The first two are pure X cases, meaning that they live in $C_3(A)$; the last two are hybrid cases.

Any XX.. tet dies in M_Y; any YY.. tet dies in M_X.

The remaining cases to check are the pure cases $CCCX$, $CCCY$, and the hybrid case $CCXY$. A $CCCX$ tet $t = \pm[c_1 c_2 c_3 x_1]$ dies in M_Y because $K_3(B,q)(t) = [c_1 c_2 c_3 q]$, which vanishes because $q \in C = \{c_1, c_2, c_3\}$ produces a repeated vertex. Likewise a $CCCY$ tet dies in M_X. The more interesting case is $CCXY$: The key is that since $|C| = 3$, $\{c_1, c_2\}$ cannot be disjoint from $\{p, q\}$, so it dies on one side or the other.

So

$$|M| = |M_X(p,q)| + |M_Y(p,q)|,$$

and Zvol adds. Note how we're now emphasizing the possible dependence of M_X, M_Y on p, q. When $n = 1$ the choice of p, q can indeed make a difference: We can get a different pair (M_X, M_Y) if we switch p and q. (Think about the connected sum of two cycle graphs.)

But when $n \geq 2$ we will now show that all tets are pure X or pure Y, so M_X consists of all the pure X tets, and ditto for M_Y. This will make $M = M_X + M_Y$ with $M_X \in \mathrm{taut}(X)$, $M_Y \in \mathrm{taut}(Y)$.

Again our test case is $n = 2$.

The key observation is that, now that we know that every tet dies on one side or the other, we know that no tet can die on both sides, because that would make $|M| > \mathrm{Zvol}(X) + \mathrm{Zvol}(Y)$.

An $XXYY$ tet would die on both sides, so there can be none of these.

Any $pqXY$, $pXXY$, or $qXYY$ would die on both sides, and p, q are arbitrary, so this rules out all $CCXY, CXXY, CXYY$.

The only remaining hybrids are $XXXY$ and $XYYY$. Let's rule out $XXXY$, leaving $XYYY$ to symmetry.

Fix any $y_0 \in B \setminus C$, and suppose there is some $XXXy_0$ tet in M. Let $U \subset M$ consist of all such $XXXy_0$ tets. We claim that $y_0 \notin \mathrm{Vert}(\partial U)$. For let s be any 2-simplex of the form XXy_0, the only kind of 2-simplex containing y_0 that could belong to ∂U. Any $t \in M$ of which s or $-s$ is a face must be a hybrid tet, and the only possibility is $XXXy_0$, so $t \in U$. Because s is a mixed XXy_0 face, it is not a face of $X + Y$, so its signed multiplicity in $\partial M = X + Y$ vanishes; hence it also vanishes in ∂U, because the tets of M with $\pm s$ as a face all belong to U. So $y_0 \notin \mathrm{Vert}(\partial U)$, making U a complete cone on y_0, contradicting 3. So there is no such $XXXy_0$ tet, ruling out $XXXY$, and with it $XYYY$.

So there are no hybrid tets, meaning that M splits, as claimed. $\square$

The splitting of taut fillings depends on the assumption $n \geq 2$. Let's consider what goes wrong when $n = 1$. Here the hybrid types are CXY, XXY, XYY. Neither pXY nor qXY dies on both sides, so we cannot rule out CXY. Failing that, the XXy_0 tets needn't form a complete cone, because $[x_0x_1y_0]$ can continue across $[x_1y_0]$ to $-[px_1y_0]$, so we cannot rule out XXY either.

The foregoing proof works equally well over $\mathbf{Q}$, providing we permit ourselves to work with fractional multisets. Alternatively, we can clear denominators as in the introduction above. Either way, we have:

Corollary 2. Qvol *adds under almost disjoint union, and for* $n \geq 2$ *taut* $\mathbf{Q}$*-fillings split.* $\square$

6 Triangulations

We turn now to filling simplicial triangulations of S^2. We begin by fixing the terminology.

An *n-simplex* s is a set of size $n + 1$. Its *faces* are its subsets, which are k-simplices with $-1 \leq k \leq n$, where the empty simplex has dimension -1. A *simplicial complex* σ is a finite collection of simplices closed under taking faces: if $t \subset s \in \sigma$ then $t \in \sigma$.

For any simplex $s \in \sigma$ define the *link*

$$\mathrm{link}(s, \sigma) = \{t : t \cap s = \emptyset, s \cup t \in \sigma\}.$$

This is a simplicial complex, but (except for $\mathrm{link}(\emptyset, \sigma) = \sigma$) it is not a subcomplex of σ, because we are taking simplices in σ and lowering their dimension by $\dim s + 1$.

We are interested in normal pseudomanifolds, which we will call *clean n-complexes.* A simplicial complex σ is clean of dimension n if:

1. σ is pure of dimension n, meaning that every simplex of σ is contained in some n-simplex of σ.

2. σ is a pseudomanifold: the link $\mathrm{link}(s, \sigma)$ of any $(n-1)$-simplex s consists of either one point (so s is a boundary simplex) or two points (so s is an interior simplex).

3. σ is normal: for every simplex s of dimension $0, \ldots, n-2$, the link $\mathrm{link}(s, \sigma)$ is connected.

The normality condition says that the n-simplices are glued along shared faces without additional identifications. It rules out, for example, an icosahedron with a pair of opposite vertices identified.

If σ is clean, then each link $\mathrm{link}(s, \sigma)$ is clean. The boundary $\partial\sigma$ of a clean complex is clean and closed: $\partial\partial\sigma = \{\emptyset\}$.

If W is an n-manifold, say that a simplicial complex σ is a *simplicial triangulation* of W if the geometric carrier of σ is homeomorphic to W. In this case σ is necessarily clean. When W is oriented, a choice of orientation determines one of the two associated cycles $X(\sigma) \in C_n$; our arguments do not depend on which of the two orientations is chosen.

This notion of triangulation is not as general as one might want for some purposes. The double cover of a triangle is not a simplicial triangulation of S^2, because that would require two distinct 2-simplices to have the same three edges. Nor can any 2-simplex have two of its edges glued to one another. Thurston [6] allows such triangulations, but it is unclear how important these are to his theory of shapes of surfaces. We do not allow them here.

For a chain $M \in C_n$, define $K(M)$ to be the pure simplicial complex generated by the unoriented n-simplices on which M has nonzero coefficient. We call M *simplicial* if every nonzero coefficient is ± 1, so no unoriented n-simplex is repeated. We call M *clean* if M is simplicial and $K(M)$ is a clean simplicial complex.

7 Filling a triangulation of the 2-sphere

Theorem 2. *If σ is a simplicial triangulation of S^2 and M is a taut filling of $X(\sigma)$, then M is clean and $K(M)$ is a simplicial triangulation of B^3.*

Proof. Call (σ, M) a *filling pair* if σ is a simplicial triangulation of S^2 and M is a taut filling of $X(\sigma)$. Call a filling pair *good* if M is clean and $K(M)$ is a simplicial triangulation of B^3. We will show that every filling pair is good.

Suppose not, and choose a bad filling pair (σ, M) with $|M|$ minimal. Let v, e, f count the vertices, edges, and faces of σ. Then $e = 3v - 6$ and $f = 2v - 4$. The tetrahedral boundary is good, so $v > 4$. Because taut fillings split under connected sum, we may assume that σ is prime, meaning not a connected sum along a triangle. In particular, σ has no vertex of degree 3.

Call a support tetrahedron t of M *eligible* if two of its oriented boundary faces are faces of $X(\sigma)$ with the correct signs. It cannot have three such faces, since that would give a degree-3 vertex in σ. By Proposition 2,

$$|M| = \mathrm{Zvol}(\sigma) \leq f - \mathrm{maxdeg}(\sigma).$$

For each support tetrahedron u of M, let $A(u)$ be the set of faces of σ that occur as correctly oriented boundary faces of u. Each $A(u)$ has size at most 2. Every face of σ occurs in at least one $A(u)$, because its coefficient in $\partial M = X(\sigma)$ is nonzero with the prescribed sign. List the support tetrahedra of M and reveal the sets $A(u)$ one at a time, recording only faces of σ not previously seen. If r support tetrahedra reveal two new faces, then

$$f \leq |\operatorname{supp} M| + r \leq |M| + r.$$

Thus $r \geq \mathrm{maxdeg}(\sigma)$. Consequently there are at least $\mathrm{maxdeg}(\sigma)$ eligible support tetrahedra whose pairs of boundary faces are mutually disjoint. We will use only the fact that, given any boundary face s and any eligible tet t, there is another eligible tet whose boundary faces are disjoint from those of t and do not include s.

Let t be an eligible tet with boundary faces $s_1 = [a, b, c]$ and $s_2 = [b, a, d]$. Removing t from M gives a taut filling $M - t$ of the cycle obtained from $X(\sigma)$ by replacing s_1, s_2 with $[c, d, b]$ and $[d, c, a]$. At the level of the boundary complex, this flips the edge ab to cd. There are two cases.

First, if cd was not an edge of σ, the flipped complex σ_t is again a simplicial triangulation of S^2. By minimality, $(\sigma_t, M - t)$ is good. Adding t back gives a triangulation of B^3 unless the same support tetrahedron t was already present in $M - t$, i.e. unless t had multiplicity 2 in M. If that happened, remove instead an eligible tet u whose boundary faces are disjoint from those of t. Then $M - u$ would still contain the repeated support tetrahedron t and could not be simplicial, so $(\sigma_u, M-u)$ would be a smaller bad filling pair, a contradiction.

Second, if cd was already an edge of σ, the flipped cycle is an almost disjoint union of two simplicial triangulations σ_1, σ_2 of S^2, conjoined along the edge cd. By Theorem 1, $M-t$ splits as M_1+M_2, where M_i is a taut filling of $X(\sigma_i)$. By minimality each (σ_i, M_i) is good. The complex $K(M-t)$ need not be clean as a single complex, because the link of the shared edge cd can be disconnected; nevertheless it is the union of the two clean pieces $K(M_1)$ and $K(M_2)$. Gluing these two triangulated balls to the two sides of t gives a triangulated ball, unless t already occurred in $M-t$. The same disjoint eligible-tet argument rules out that multiplicity-2 obstruction.

The preceding paragraph shows that, for an eligible tet t, the intended reassembly is legitimate unless a local obstruction was already present in M away from t. Such obstructions are also excluded by the same minimality argument, using the supply of disjoint eligible tets. The point is that an obstruction to cleanness is local: it is witnessed by one repeated tet, by one triangle with too many incident tets, or by one disconnected link. If we choose an eligible tet whose two boundary faces are disjoint from the finitely many boundary faces needed to change that local witness, then the witness persists after the flip; for a repeated tetrahedron, avoid the boundary faces of that tetrahedron; for an over-incident triangle, avoid the boundary faces through which the incidence could change; and for a disconnected link, avoid the unique tetrahedron whose removal would bridge the exceptional component. In the one-sphere case it persists in the smaller filling $M-u$; in the conjoined case it persists in one of the two split summands, unless it is exactly the artificial disconnected link of the new common edge. That exceptional disconnected link is the reason we split $M-u$ into two pieces before applying induction.

We now spell this out for the clean-complex conditions.

1. M has no repeated tetrahedron. If the same unoriented tet occurred twice in M, then removing an eligible tet u not equal to one of these copies and not using any of the relevant boundary faces would leave the repeated tet in the smaller filling, or in one of the two smaller fillings after a conjoined split. This would contradict minimality, since the smaller filling could not be clean.

2. $K(M)$ is a pseudomanifold. Suppose a 2-simplex s occurs as a face of more than two support tetrahedra of M. If $s \notin \sigma$, the excess is an

interior incidence; removing an eligible tet avoiding the tets accounting for this incidence leaves the same excess in the smaller one-sphere filling or in one of the two split pieces. If $s \in \sigma$, choose the eligible tet so that s is not one of its two boundary faces; the boundary excess again persists. Either way we get a smaller bad filling pair, contradicting minimality.

3. No edge $\{a, b\}$ has disconnected link in $K(M)$. The link is a 1-dimensional complex whose edges correspond to support tetrahedra of M containing $\{a, b\}$. If a component of such a link has fewer than three vertices, then it is a single edge $\{c, d\}$, corresponding to a tet $\{a, b, c, d\}$. In that case $\{a, b\}$ is a boundary edge of σ, and $\{a, b, c\}$ and $\{a, b, d\}$ are the two boundary faces adjacent along it. There can be at most one such isolated-edge component. If the link were disconnected, remove an eligible tet away from this exceptional component. The disconnected link remains in the one-sphere case or in one of the split summands in the conjoined case, contradicting minimality.

4. No vertex has disconnected link in $K(M)$. If $\mathrm{link}(\{a\}, K(M))$ were disconnected, the only way a single removal could connect it would be for one component of the link to be a single 2-simplex $\{b, c, d\}$ and for the removed tet to be $\{a, b, c, d\}$. Then $\{a, b, c\}$, $\{a, b, d\}$, and $\{a, c, d\}$ would all be boundary faces of σ, so a would have degree 3 in σ, contrary to primality. Thus a disconnected vertex link would persist under a suitable smaller removal and again contradict minimality.

Thus M is simplicial, $K(M)$ is clean, and $K(M)$ is a simplicial triangulation of B^3, contradicting that (σ, M) was bad. $\square$

After Theorem 2 we will often identify a taut filling M of $X(\sigma)$ with the triangulated ball $K(M)$.

8 Shelling

A *shelling* of a simplicial triangulation τ of B^3 is an ordering $(t_1, \ldots, t_{|\tau|})$ of its tetrahedra such that, for each k, the subcomplex τ_k generated by $\{t_1, \ldots, t_k\}$ is a simplicial triangulation of B^3. For $k > 1$, the new tet may meet τ_{k-1} in one, two, or three faces. Gluing along one face increases the boundary vertex count by 1; gluing along two faces leaves it unchanged; gluing along three faces would bury an interior vertex, and so cannot occur in the shellings obtained below. Thus the boundary vertex count never decreases. Call τ *freely shellable* if any tet of τ can be chosen as the first tet of such a shelling.

Theorem 3. *If σ is a simplicial triangulation of S^2 and M is a taut filling of $X(\sigma)$, then $K(M)$ is a freely shellable simplicial triangulation of B^3.*

Proof. By Theorem 2, $K(M)$ is a triangulated ball; write $\tau = K(M)$. Fix a tet $t_1 \in \tau$ that we want to remove last. We describe a recursive procedure, called *shucking*, that takes τ apart so as to remove t_1 last; reversing the shucking order gives the shelling.

If τ consists only of t_1, there is nothing to prove. If τ has a boundary vertex of degree 3 lying in a tet $t \neq t_1$, remove that tet and continue with $\tau - t$. If every degree-3 boundary vertex belongs to t_1, let t_2 be the unique tet of $\tau - t_1$ attached to t_1 across the opposite face; shuck $\tau - t_1$ ending with t_2, and then remove t_1.

It remains to consider the case where the boundary has no degree-3 vertex. As in the proof of Theorem 2, there are at least $\mathrm{maxdeg}(\partial\tau) \geq 4$ disjointly eligible tets. Choose one, say t, whose two boundary faces are disjoint from those of t_1. If removing t leaves one triangulated ball, remove t and continue ending with t_1. If removing t splits the boundary into two conjoined spheres, then $\tau - t$ splits into two triangulated balls τ_1, τ_2 by Theorem 1 and Theorem 2. Assume $t_1 \in \tau_1$, and let t_2 be a tet of τ_2 adjacent to the gluing face of t. Shuck τ_2 ending with t_2, remove t, and then shuck τ_1 ending with t_1. Every removal is the reverse of attaching along one or two faces, so the reversed order is the desired shelling. □

9 Flag complex

A simplicial complex is a *flag complex* (or clique complex) if every clique in its 1-skeleton occurs as a simplex. A simplicial triangulation σ of S^2 is flag exactly when it is prime (not the connected sum of smaller complexes) and is not the boundary of a tetrahedron.

We can decompose any σ as the connected sum of prime components σ_i. From the point of view of taut fillings these components are independent: any taut filling of $X(\sigma)$ is the union of taut fillings of the $X(\sigma_i)$. We now prove that, regardless of whether σ itself is prime, every taut filling is flag.

Theorem 4. *If σ is a simplicial triangulation of S^2 and M is a taut filling of $X(\sigma)$, then $K(M)$ is a flag complex.*

Proof. Let $\tau = K(M)$. We must rule out the following minimal failures of flagness:

1. the three edges of a triangle $\{A, B, C\}$ without the face ABC;

2. the six edges of the K_4 on $\{A, B, C, D\}$ without the tet $ABCD$;

3. the ten edges of the K_5 on $\{A, B, C, D, E\}$.

If Configurations 1 and 2 are absent, so is Configuration 3. Suppose the five vertices $\{A, B, C, D, E\}$ carried all ten edges. With no empty K_4, all five tetrahedra on four of these vertices are simplices of τ; let $U \subset M$ be the subchain formed by those five tetrahedra. Each of the ten triangles on the five vertices is a face of exactly two of the five tetrahedra, and by the pseudomanifold condition from Theorem 2 these are the only tetrahedra of M containing it. Thus the triangle is interior, so it is not a face of σ and has coefficient 0 in $\partial M = X(\sigma)$. Hence ∂U vanishes on every triangle, so $\partial U = 0$. By Corollary 1 this forces $U = 0$, contradicting that U contains five tetrahedra.

We rule out Configurations 1 and 2 by minimal counterexample. If σ has a degree-3 vertex, then σ is a connected sum with one summand the tetrahedral boundary; removing that summand from σ and the corresponding tetrahedron from τ gives a smaller counterexample. So assume σ has no degree-3 vertex. By the eligible-tet count of Theorem 2 there are then at least $\mathrm{maxdeg}(\sigma) \geq 4$ eligible tetrahedra with pairwise disjoint pairs of boundary faces; in particular their flip edges are distinct, since the two boundary faces sharing an edge are the only two faces of σ along it. Removing an eligible tet flips its edge of σ, leaving either a smaller triangulated sphere or two spheres conjoined along the new edge. In the conjoined case a forbidden K_3 or K_4, all of whose vertices are joined by edges, lies on one side, where it gives a smaller counterexample.

Configuration 1. Removing an eligible tet deletes from τ only the flipped edge. The eligible tets have distinct flip edges, and the forbidden triangle has only three edges, so among at least four eligible tets some flip edge avoids all three; removing that tet preserves the triangle, a contradiction.

Configuration 2. Assume Configuration 1 is already ruled out, so the six edges on $\{A, B, C, D\}$ make all four triangles ABC, ABD, ACD, BCD

simplices of τ. We claim that removing *any* eligible tet t leaves the empty K_4 in place, so a single removal already gives a smaller counterexample. Removal creates no tetrahedron, so $\{A, B, C, D\}$ is still not one. Each of the six edges survives: for an edge, say AB, the triangles ABC and ABD are simplices, and t contains at most one of them, since containing both would force $\{A, B, C, D\} \subseteq t$, that is $t = \{A, B, C, D\}$, which is not a tetrahedron. So one of them, say ABD, is not contained in t; being a simplex it is a face of some tetrahedron $u \neq t$, and $AB \subseteq ABD \subseteq u$ shows AB survives the removal of t. Hence all six edges remain and the empty K_4 persists, giving a smaller counterexample. No maximum-degree bound and no octahedral special case is needed. $\square$

References

[1] Peter Doyle, Matthew Ellison, and Zili Wang, *Filling a triangulation of the 2-sphere*, 2024, `https://doi.org/10.48550/arXiv.2303.10773`.

[2] Matthew Ellison, *Lower bounds and integrality gaps in simplicial decomposition*, 2024, `https://doi.org/10.48550/arXiv.2404.01279`.

[3] Claire Mathieu and William Thurston, *Rotation distance using flows*, 1992, `https://doi.org/10.48550/arXiv.2409.17905`.

[4] Lionel Pournin and Zili Wang, *Strong convexity in flip-graphs*, 2021, `https://doi.org/10.48550/arXiv.2106.08012`.

[5] Daniel D. Sleator, Robert E. Tarjan, and William P. Thurston, *Rotation distance, triangulations, and hyperbolic geometry*, J. Amer. Math. Soc. **1** (1988), 647–681, `https://doi.org/10.2307/1990951`.

[6] William P. Thurston, *Shapes of polyhedra and triangulations of the sphere*, 1998, `https://doi.org/10.48550/arXiv.math/9801088`.